Customer Churn Analysis and Prediction

Power BI Internship Project Documentation

Organization	Saiket Systems
Tool	Microsoft Power BI
Dataset	Telco Customer Churn Dataset
Records	7,043 customers

1. Project Objective

The project analyzes customer churn in a telecommunications dataset and uses Power BI to present churn patterns, customer demographics, tenure, contract type, payment method, and internet service. The objective is to identify factors associated with customer churn and provide actionable recommendations to improve customer retention.

Project scope completed

- Import the provided customer churn dataset into Power BI.
- Clean and transform the data for analysis and visualization.
- Create DAX measures for customer and churn KPIs.
- Build a professional churn analysis dashboard.
- Analyze customer demographics, tenure, contract, payment method, and internet service.
- Summarize business insights and recommendations.

2. Dataset Overview

Attribute	Details
Rows	7,043
Columns	21
Target variable	Churn (Yes / No)
Churned customers	1,869
Active customers	5,174
Overall churn rate	26.54%
Tenure range	0–72 months
Average tenure	32.37 months
Missing TotalCharges	11 records

3. Data Connection & Transformation

The customer churn CSV dataset was imported into Power BI through the Text/CSV connector and opened in Power Query for preprocessing.

Cleaning and transformation performed

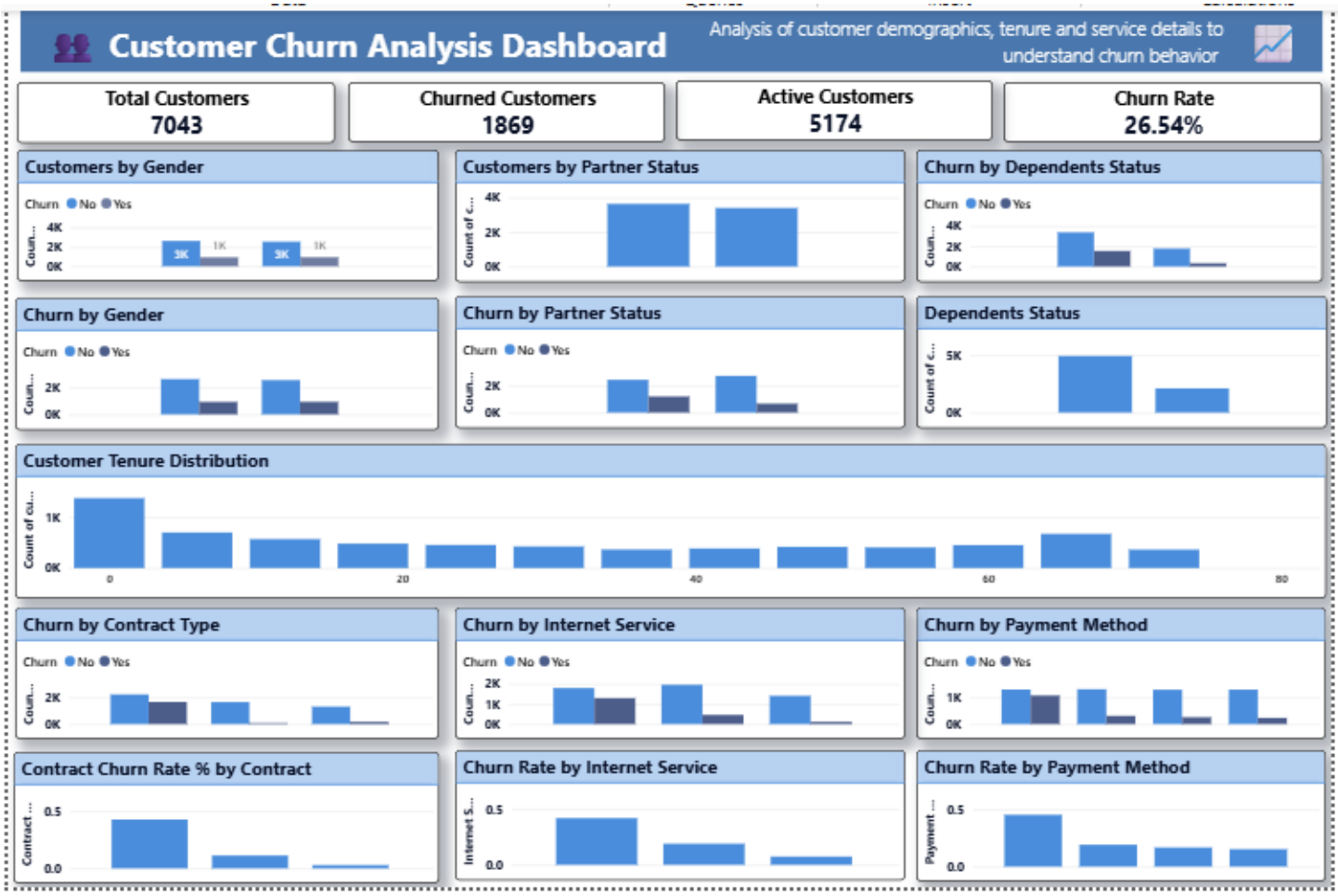
- Checked the dataset structure and data types.
- Converted tenure to Whole Number.
- Converted MonthlyCharges to Decimal Number.
- Converted TotalCharges to Decimal Number.
- Kept categorical fields such as gender, contract, payment method and internet service as text.
- Handled the 11 blank TotalCharges records as null values; these records correspond to customers with zero-month tenure.
- Applied the transformations and loaded the cleaned data into the Power BI model.

4. DAX Measures

Measure	DAX Logic
Total Customers	COUNTROWS('Telco Customer Churn')
Churned Customers	CALCULATE(COUNTROWS('Telco Customer Churn'), 'Telco Customer Churn'[Churn] = "Yes")
Active Customers	CALCULATE(COUNTROWS('Telco Customer Churn'), 'Telco Customer Churn'[Churn] = "No")
Churn Rate %	DIVIDE([Churned Customers], [Total Customers], 0)

5. Power BI Dashboard

The final dashboard combines KPI cards, demographic analysis, tenure distribution, churn analysis by service attributes, and churn-rate comparisons.



Final dashboard view

Dashboard components

- KPI overview: Total Customers, Churned Customers, Active Customers and Churn Rate.
- Customer demographics: Gender, Partner Status and Dependents Status.
- Customer tenure distribution in months.
- Churn by Contract Type, Internet Service and Payment Method.
- Churn-rate comparisons for Contract Type, Internet Service and Payment Method.

6. Key Insights

The dashboard contains 7,043 customers. 1,869 customers have churned and 5,174 remain active, resulting in an overall churn rate of 26.54%.

- **Contract:** Month-to-month has the highest churn rate (42.7%). Two-year contracts have the lowest churn (2.8%).
- **Payment method:** Electronic check has the highest churn (45.3%). Automatic credit-card payment has a lower churn rate (15.2%).
- **Internet service:** Fiber optic customers show the highest churn (41.9%), while customers with no internet service show the lowest (7.4%).
- **Partner status:** Customers without a partner have higher churn (33.0%) than customers with a partner (19.7%).
- **Dependents:** Customers without dependents have higher churn (31.3%) than customers with dependents (15.5%).
- **Tenure:** Customer tenure ranges from 0 to 72 months, with an average of 32.4 months.

7. Business Recommendations

- **Reduce month-to-month churn —** Offer incentives for customers to move from month-to-month plans to one-year or two-year contracts, such as loyalty discounts, bundled services, or upgrade benefits.
- **Address electronic-check churn —** Investigate payment experience and encourage customers to adopt automatic bank transfer or credit-card payments where appropriate.
- **Improve fiber-optic retention —** Review service quality, pricing, support response times and customer experience for fiber-optic users.
- **Focus on early-tenure customers —** Create an onboarding and early-retention program for newer customers, including proactive support and satisfaction checks.
- **Use targeted retention campaigns —** Prioritize high-risk combinations such as month-to-month contract + fiber optic + electronic check for proactive retention outreach.

8. Conclusion

The Power BI dashboard provides a consolidated view of customer churn and the factors associated with it. The analysis highlights contract type, payment method, internet service, tenure and customer demographics as useful dimensions for retention planning. The dashboard can support business teams in identifying higher-risk customer segments and designing targeted retention actions.

This documentation is prepared for the Saiket Systems Power BI Internship project.